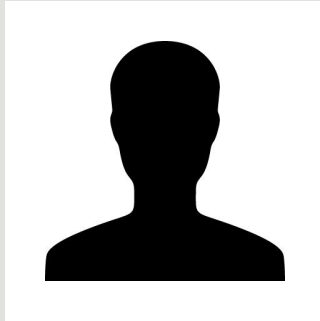


Quantum Computing

Can quantum computing go up against
conventional computing?

Presenters



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What is computing?

Computing is the process of using computer technology to complete a given goal-oriented task.

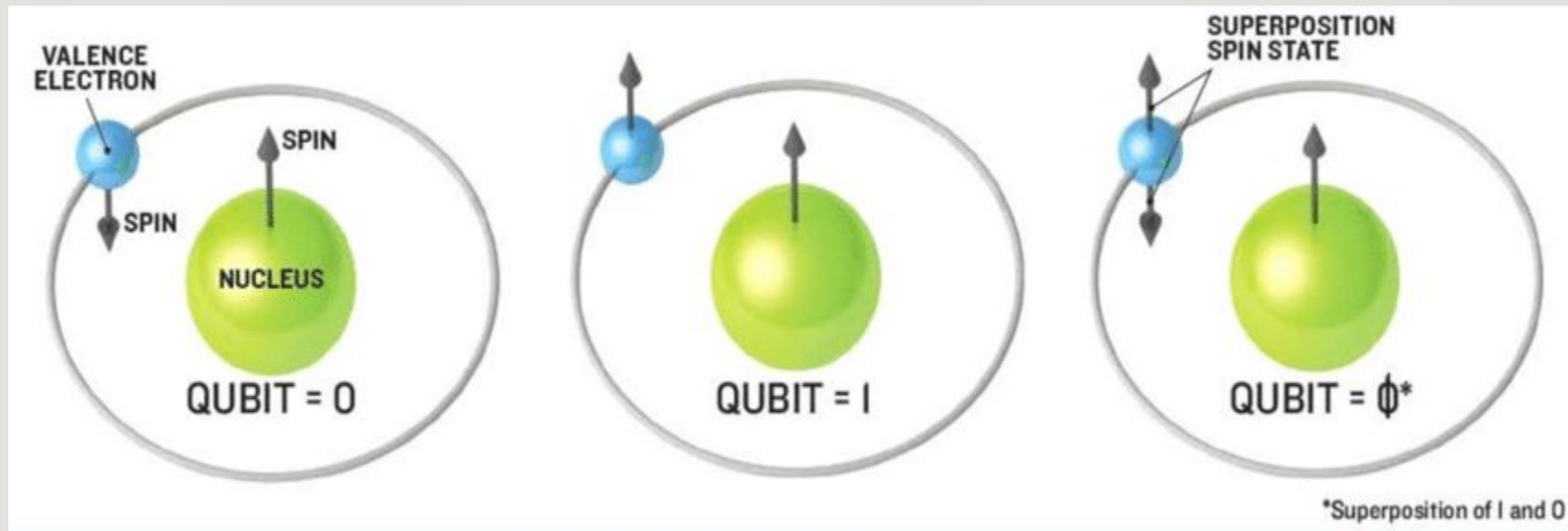
Computing has also been defined as a branch of engineering science that deals with the systematic study of algorithmic processes, which are used to describe and transform information.

Quantum Computing

It's not using zeros and ones like classical computers are – bits and bytes – but it is actually able to work with something called qubits.

Qubits are quantum bits, and have the special property that at the same time they can be zero and one.

The quantum bits can be on and off at the same time.



The effects of superposition

As the superposition state consists of many different states at once, it allows the user to probe many possibilities at a given point in time.

An example would be; you are working in finance and you want to say which portfolio has the largest profit, you need to take many, many different cases into account and then find the best one.

There would be various variables at play, and a quantum computer would allow you to account for these multiple variables.

Current Development

Development is gradual, however given the fact that majority of the quantum physics was developed in Europe over the last century, one can assume development is in demand. This is according to Dr Thomas Munz, an academic who is leading a study in quantum computing.

He states,

There are currently several proof of concept implementations. Companies are working in the finance area on portfolio optimisation. Certain companies are working towards chemistry – one prominent example is how to generate fertiliser.

Shor's Algorithm

Although any integer number has a unique decomposition into a product of primes, finding the prime factors is believed to be a hard problem.

This assumption was challenged in 1995 when Peter Shor proposed a polynomial-time quantum algorithm for the factoring problem.

In fact, Shor's algorithm can be run using a computing computer with plenty of qubits, and can later be modified to serve as an effective algorithm for quantum computers.

Promising Research

UNIVERSITAET INNSBRUCK in Austria has been conducting the study named AQTION (Advanced quantum computing with trapped ions).

It is being funded by the European body CORDIS. The study produced several papers linked below.

<https://cordis.europa.eu/project/id/820495>

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